Experiment - 03

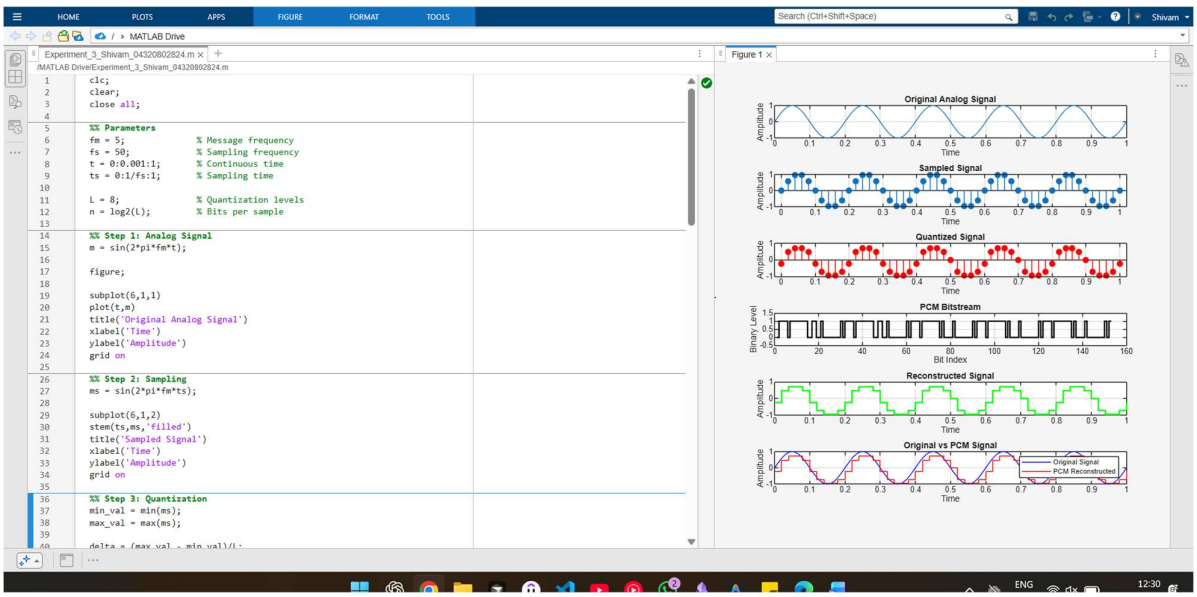
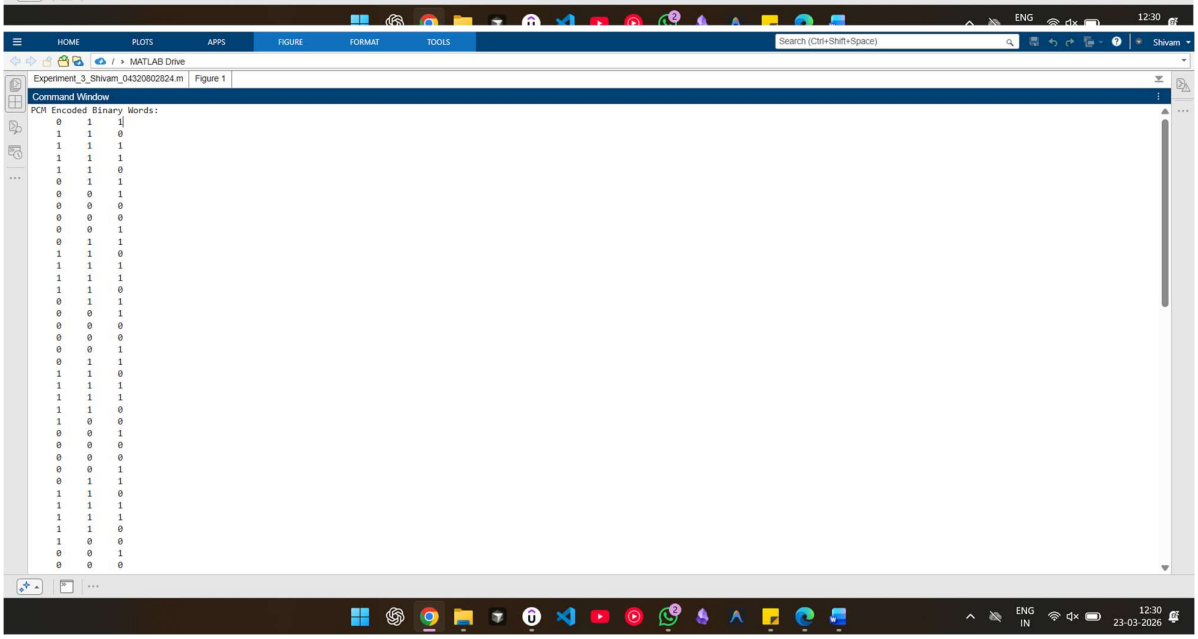
AIM: To perform Pulse Code Modulation (PCM) using MATLAB

Software used: MATLAB

Code:

```
clc;
clear;
close all;
%% Parameters
fm = 5;             % Message frequency
fs = 50;            % Sampling frequency
t = 0:0.001:1;      % Continuous time
ts = 0:1/fs:1;      % Sampling time
L = 8;              % Quantization levels
n = log2(L);        % Bits per sample
%% Step 1: Analog Signal
m = sin(2*pi*fm*t);
figure;
subplot(6,1,1)
plot(t,m)
title('Original Analog Signal')
xlabel('Time')
ylabel('Amplitude')
grid on
%% Step 2: Sampling
ms = sin(2*pi*fm*ts);
subplot(6,1,2)
stem(ts,ms,'filled')
title('Sampled Signal')
xlabel('Time')
ylabel('Amplitude')
grid on
```

```
%% Step 3: Quantization
min_val = min(ms);
max_val = max(ms);
delta = (max_val - min_val)/L;
index = floor((ms - min_val)/delta);
index(index >= L) = L-1;   % Prevent overflow
q = min_val + index*delta;
subplot(6,1,3)
stem(ts,q,'r','filled')
title('Quantized Signal')
xlabel('Time')
ylabel('Amplitude')
grid on
%% Step 4: PCM Encoding
encoded = de2bi(index,n,'left-msb');
disp('PCM Encoded Binary Words:')
disp(encoded)
%% Step 5: Convert to Bitstream
bitstream = reshape(encoded',1,[]);
subplot(6,1,4)
stairs(bitstream,'k','LineWidth',1.5)
ylim([-0.5 1.5])
title('PCM Bitstream')
xlabel('Bit Index')
ylabel('Binary Level')
grid on
%% Step 6: Reconstructed Signal
subplot(6,1,5)
stairs(ts,q,'g','LineWidth',1.5)
title('Reconstructed Signal')
xlabel('Time')
ylabel('Amplitude')
grid on
%% Step 7: Original vs PCM
```

```
subplot(6,1,6)
plot(t,m,'b')
hold on
stairs(ts,q,'r')
legend('Original Signal','PCM Reconstructed')
title('Original vs PCM Signal')
xlabel('Time')
ylabel('Amplitude')
grid on
```

Output: